

Probabilistic Reachability Bounds for Multi-Phase Entry under Process and Measurement Noise

Ethan Knox

Independent Researcher, Champaign, IL, USA

2026

Abstract

Keywords: stochastic reachability; Fokker-Planck equation; occupation measures; supermartingales; stochastic barrier certificates; concentration of measure; hybrid systems; atmospheric entry

MSC 2020:

Contents

1	Introduction	5
1.1	The price of hard bounds	5
1.2	Process noise and measurement noise	5
1.3	Thesis statement	5
1.4	Outline of contributions	5
2	The Stochastic Multi-Phase Model	6
2.1	Ito dynamics per mode	6
2.2	Where the diffusion comes from	6
2.3	Guards under noise: hitting times and boundary behavior	6
2.4	The observation model with unbounded noise	6
3	From Liouville to Fokker-Planck	7
3.1	The stochastic occupation measure	7
3.2	The generator and the weak forward equation	7
3.3	Per-mode measures and stochastic linkage	7
4	Concentration Against Diffusion	8
4.1	Two competing limits	8
4.2	Diffusive spreading of the boundary layers	8
4.3	The critical noise scale	8
4.4	Where the real atmosphere sits	8
4.5	Consequence for the certificate	8
5	Probabilistic Certificates	9
5.1	Why containment is unavailable	9
5.2	Supermartingales and the exit probability	9
5.3	Deriving the certificate	9
5.4	Numerical cross-check by sum-of-squares	9
5.5	Trading probability against volume	9
5.6	Set erosion and probabilistic tubes	9
6	Sequential Conditioning under Unbounded Noise	10
6.1	From restriction to reweighting	10
6.2	The conditioned stochastic occupation measure	10
6.3	Contraction in probability	10
6.4	Process noise against measurement information	10
7	Verification and Results	11
7.1	Recovering the deterministic limit	11
7.2	Where the real atmosphere sits	11
7.3	Exit probability against Monte Carlo	11
7.4	Probability against volume	11
7.5	Contraction under noisy returns	11
7.6	The three-way balance	11
7.7	Reproducibility environment	11

8	Discussion	12
8.1	Limitations	12
8.2	What the three papers establish together	12
8.3	Extensions	12
A	Ito Calculus Across Guards	14
A.1	The generator of the hybrid diffusion	14
A.2	Hitting times and the boundary	14
A.3	The deterministic limit	14
B	Diffusion Models from Atmospheric Data	14
B.1	Density variability	14
B.2	Wind variability	14
B.3	Polynomial form of the diffusion coefficient	14
C	Supermartingale Inequalities	14
C.1	The maximal inequality	14
C.2	Finite-horizon and stopped versions	14
C.3	Composition over epochs	14
D	Proof of the Concentration-Diffusion Threshold	14
D.1	Widths of the two effects	14
D.2	The threshold	14
D.3	The inflated residual	14

Open items

conditional on both prior papers (paper 1 for the embedding and the profile decomposition, paper 2 for the conditioning step).	6
make-or-break condition for this paper: if measured density and wind variability put the real atmosphere above the critical noise scale, diffusion washes out the boundary layers and there is no skeleton left to bound around (paper 1's decomposition would not survive contact with noise).	8

1 Introduction

1.1 The price of hard bounds

1.2 Process noise and measurement noise

1.3 Thesis statement

1.4 Outline of contributions

2 The Stochastic Multi-Phase Model

2.1 Ito dynamics per mode

conditional on both prior papers (paper 1 for the embedding and the profile decomposition, paper 2 for the conditioning step).

2.2 Where the diffusion comes from

2.3 Guards under noise: hitting times and boundary behavior

2.4 The observation model with unbounded noise

3 From Liouville to Fokker-Planck

3.1 The stochastic occupation measure

3.2 The generator and the weak forward equation

3.3 Per-mode measures and stochastic linkage

4 Concentration Against Diffusion

4.1 Two competing limits

4.2 Diffusive spreading of the boundary layers

4.3 The critical noise scale

4.4 Where the real atmosphere sits

make-or-break condition for this paper: if measured density and wind variability put the real atmosphere above the critical noise scale, diffusion washes out the boundary layers and there is no skeleton left to bound around (paper 1's decomposition would not survive contact with noise).

4.5 Consequence for the certificate

5 Probabilistic Certificates

- 5.1 Why containment is unavailable
- 5.2 Supermartingales and the exit probability
- 5.3 Deriving the certificate
- 5.4 Numerical cross-check by sum-of-squares
- 5.5 Trading probability against volume
- 5.6 Set erosion and probabilistic tubes

6 Sequential Conditioning under Unbounded Noise

6.1 From restriction to reweighting

6.2 The conditioned stochastic occupation measure

6.3 Contraction in probability

6.4 Process noise against measurement information

7 Verification and Results

- 7.1 Recovering the deterministic limit
- 7.2 Where the real atmosphere sits
- 7.3 Exit probability against Monte Carlo
- 7.4 Probability against volume
- 7.5 Contraction under noisy returns
- 7.6 The three-way balance
- 7.7 Reproducibility environment

8 Discussion

8.1 Limitations

8.2 What the three papers establish together

8.3 Extensions

References

A Ito Calculus Across Guards

A.1 The generator of the hybrid diffusion

A.2 Hitting times and the boundary

A.3 The deterministic limit

B Diffusion Models from Atmospheric Data

B.1 Density variability

B.2 Wind variability

B.3 Polynomial form of the diffusion coefficient

C Supermartingale Inequalities

C.1 The maximal inequality

C.2 Finite-horizon and stopped versions

C.3 Composition over epochs

D Proof of the Concentration-Diffusion Threshold

D.1 Widths of the two effects

D.2 The threshold

D.3 The inflated residual